

Probability
Chapter 2 - Module 2

Joint Random Variables

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June 21, 2026

In our study of probability we will need to work with many random variables at once. For example, if $X_1, X_2, \dots, X_n$ are n random variables, we will often be interested in $Y = \sum_{k=1}^n X_k$, which is itself a random variable.

We begin with some definitions focusing first on just two random variables.

Def. (CDF) Let X and Y be two random variables. We define the CDF for X and Y (or equivalently for (X, Y)) as

$$F(a, b) = P(X \leq a \wedge Y \leq b) = P(X \leq a, Y \leq b) = \sum_{\substack{x \leq a, y \leq b \\ p(x, y) > 0}} p(x, y).$$

Notation. Since we have two random variables we will write

$$F_X(a) = P(X \leq a) \quad \text{and} \quad F_Y(b) = P(Y \leq b).$$

For discrete random variable X and Y we will write

$$p(x, y) = P(X = x, Y = y),$$

where $p(x, y)$ is positive at mass points, which are points where $P(X = x, Y = y) > 0$.

Then we define the marginal mass points functions as

$$p_X(x) = \sum_{y|p(x,y)>0} p(x, y) \quad \text{and} \quad p_Y(y) = \sum_{x|p(x,y)>0} p(x, y).$$

In this way, the mass function for (X, Y) defines the mass functions for X and Y .

For continuous random variables X and Y , there is a joint probability density function $f(x, y)$ so that for any set $D \subset \mathcal{R}^2$,

$$P((X, Y) \in D) = \int_D \int f(x, y) dy dx$$

and where

$$F(a, b) = P(X \leq a, Y \leq b) = \int_{-\infty}^a \int_{-\infty}^b f(x, y) dy dx.$$

Similar to the discrete formulas,

$$\begin{aligned} P(X \leq a) &= P(X \leq a, Y \in \mathcal{R}) \\ &= \int_{-\infty}^a \left(\int_{-\infty}^{\infty} f(x, y) dy \right) dx \\ &= \int_{-\infty}^a f_X(x) dx, \end{aligned}$$

where

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy.$$

Similarly

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx.$$

When two discrete random variables are independent, then by definition

$$P(X \in A, Y \in B) = P(X \in A)P(Y \in B)$$

for all sets A and B .

And so in particular

$$P(X = x, Y = y) = P(X = x)P(Y = y).$$

This immediately translates to

$$p(x, y) = p_X(x)p_Y(y)$$

and this is a necessary and sufficient condition for the random variables X and Y to be independent.

Can we use this analysis for two continuous random variables that are independent?

Suppose that X and Y are continuous random variables. Then for every a and b we have that

$$P(X = a, Y = b) = P(X = a)P(Y = b).$$

What does this tell us?

Unfortunately, nothing since

$$P(X = a) = \int_a^a f_X(x) dx = 0,$$

$$P(Y = b) = \int_b^b f_Y(y) dy = 0$$

and

$$P(X = a, Y = b) = \int_a^a \int_b^b f(x, y) dy dx = 0.$$

Let's try something else.

Suppose that X and Y are independent continuous random variables. Then for any real numbers a and b ,

$$P(X \leq a, Y \leq b) = P(X \leq a)P(Y \leq b).$$

So let $f(x, y)$ be the pdf for (X, Y) and $f_X(x)$ and $f_Y(y)$ be the pdf's for X and Y respectively. Then

$$\int_{-\infty}^a \int_{-\infty}^b f(x, y) dy dx = \int_{-\infty}^a f_X(x) dx \int_{-\infty}^b f_Y(y) dy$$

for every a and b . So then

$$\int_{-\infty}^a \int_{-\infty}^b f(x, y) - f_X(x)f_Y(y) dy dx = 0$$

for every a and b . Then clearly

$$f(x, y) - f_X(x)f_Y(y) = 0$$

and so

$$f(x, y) = f_X(x)f_Y(y).$$

Let

$$f(x) = \begin{cases} 1 & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases} \quad \text{and} \quad g(x) = \begin{cases} 1 & 0 < x < 1 \\ 0 & \text{otherwise.} \end{cases}$$

Then

$$\int_c^d f(x) dx = \int_c^d g(x) dx$$

for any choice of c and d even though $f \neq g$.

Suppose that we have two pdf's f and g and we know that for ever c and d

$$\int_c^d f(x) dx = \int_c^d g(x) dx.$$

If f and g are continuous, then $f = g$ everywhere. Otherwise $f(x) = g(x)$ except on a null set which is also known as a set of measure zero. We will not pursue this further, but you can find more about this in any advanced calculus book.

Returning to our argument. We note that if X and Y are independent then

$$\int_{-\infty}^a \int_{-\infty}^b f(x, y) - f_X(x)f_Y(y) dy dx = 0$$

for every a and b . Then

$$f(x, y) = f_X(x)f_Y(y)$$

except on a null set.

The arguments are easily reversed to show that if

$$f(x, y) = f_X(x)f_Y(y),$$

then X and Y are independent.

Theorem. Two continuous random variables are independent if and only if

$$f(x, y) = f_X(x)f_Y(y)$$

except on a null set.

We will frequently want to analyze several random variables. The following theorem will be useful.

Theorem. Let $X_1, X_2, \dots, X_n$ be n random variables with pdf $f_{X_i}(x)$ for $1, 2, \dots, n$ and with joint pdf $f(x_1, x_2, \dots, x_n)$. Then $X_1, X_2, \dots, X_n$ are independent if and only if

$$f(x_1, x_2, \dots, x_n) = f_{X_1}(x_1) f_{X_2}(x_2) \cdots f_{X_n}(x_n)$$

except perhaps on a null set.

Note that, for example

$$f_{X_1}(x_1) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(x_1, x_2, \dots, x_n) dx_n \cdots dx_3 dx_2.$$

Theorem. Let $X_1, X_2, \dots, X_n$ be independent and let $g_i : \mathcal{R} \rightarrow \mathcal{R}$ for $i = 1, \dots, n$. Then $g_1(X_1), g_2(X_2), \dots, g_n(X_n)$ are independent.

Proof. We prove this in the case that $n = 2$ and that X and Y are independent continuous random variables. And since X and Y are independent, let the joint density function be

$$f(x, y) = f_X(x)f_Y(y).$$

Then

$$\begin{aligned} P(g_1(X) \in A \wedge g_2(Y) \in B) &= \int \int_{g_1(x) \in A \wedge g_2(y) \in B} f(x, y) dy dx \\ &= \int_{g_1(x) \in A} \int_{g_2(y) \in B} f_X(x)f_Y(y) dy dx \\ &= \int_{g_1(x) \in A} f_X(x) dx \int_{g_2(y) \in B} f_Y(y) dy \\ &= P(g_1(X) \in A) P(g_2(Y) \in B). \end{aligned}$$

Therefore $g_1(X)$ and $g_2(Y)$ are independent.

This proof easily extends to the general case.

Def. Let $g(x, y) : \mathcal{R}^2 \rightarrow \mathcal{R}$.

If X and Y are discrete random variable with mass function $p(x, y)$, then

$$E[g(X, Y)] = \sum_{(x, y) | p(x, y) > 0} g(x, y) p(x, y).$$

If X and Y are continuous random variables with joint pdf $f(x, y)$, then

$$E[g(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f(x, y) dy dx.$$

Let $g : \mathcal{R} \rightarrow \mathcal{R}$, but suppose that there are 2 random variables, X and Y .

In this context if $g : \mathcal{R} \rightarrow \mathcal{R}$, , what is $E[g(X)]$?

$$\begin{aligned} E_{X,Y}[g(X)] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x) f(x, y) dy dx \\ &= \int_{-\infty}^{\infty} g(x) \left(\int_{-\infty}^{\infty} f(x, y) dy \right) dx \\ &= \int_{-\infty}^{\infty} g(x) f_X(x) dx \\ &= E_X[g(X)]. \end{aligned}$$

A similar argument can be made for discrete random variables.

And similar arguments can be made for more than two random variables.

This is a relief since this implies that $E[g(X)]$ is not ambiguous and its meaning is clear and the result only depends on information about X !

Def. (Covariance) Given two random variables X and Y , we define

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])].$$

So clearly

$$\text{Var}(X) = E[(X - E[X])^2] = \text{Cov}(X, X).$$

Def. (Correlation) Given two random variables X and Y , we define

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = E \left[\frac{X - E(X)}{\sigma_X} \frac{Y - E(Y)}{\sigma_Y} \right],$$

where

$$\sigma_X = \sqrt{\text{Var}(X)} \quad \text{and} \quad \sigma_Y = \sqrt{\text{Var}(Y)}.$$

We say that two random variables X and Y are uncorrelated if and only if $\text{Cor}(X, Y) = 0$ or equivalently if $\rho(X, Y) = 0$.

Theorem. Let X , Y and Z be random variables.

Then

$$\text{Cov}(X + Y, Z) = \text{Cov}(X, Z) + \text{Cov}(Y, Z)$$

and

$$\text{Cov}(X, Y + Z) = \text{Cov}(X, Z) + \text{Cov}(X, Y).$$

Theorem. Let X and Y be random variables.

Then

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y].$$

Homework Prove these results. You may wish to use the next theorem to do this.

Theorem. Let X and Y be random variables. Then the Expectation function is linear over X and Y , meaning that for any a and b ,

$$E[aX + bY] = aE[X] + bE[Y].$$

Proof. We will prove this in the case that X and Y are continuous random variables with pdf $f_X(x)$ and $f_Y(y)$ respectively and joint pdf $f(x, y)$. Then

$$\begin{aligned} E[aX + bY] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (ax + by) f(x, y) dy dx \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} a x f(x, y) dy dx + \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} b y f(x, y) dx dy \\ &= a \int_{-\infty}^{\infty} x \int_{-\infty}^{\infty} f(x, y) dy dx + b \int_{-\infty}^{\infty} y \int_{-\infty}^{\infty} f(x, y) dx dy \\ &= a \int_{-\infty}^{\infty} x f_X(x) dx + b \int_{-\infty}^{\infty} y f_Y(y) dy \\ &= a E[X] + b E[Y]. \end{aligned}$$

Theorem. Let $X_1, X_2, \dots, X_n$ be n random variables (continuous or discrete) and let $a_1, a_2, \dots, a_n$ be real numbers. Then

$$E[a_1X_1 + a_2X_2 + \cdots a_nX_n] = a_1E[X_1] + a_2E[X_2] + \cdots a_nE[X_n].$$

Corollary. Let $X_1, X_2, \dots, X_n$ be n random variables (continuous or discrete). Then

$$E[X_1 + X_2 + \cdots X_n] = E[X_1] + E[X_2] + \cdots E[X_n].$$

Homework You should prove this in the case that $X_1, X_2, \dots, X_n$ are discrete random variables.

Question: So if X and Y are random variables, is it true that $E[XY] = E[X]E[Y]$?

Theorem. If X and Y are independent random variables, then

$$E[XY] = E[X]E[Y].$$

Proof. For a change of pace, let's prove this in the case that X and Y are discrete random variables with mass functions $p_X(x)$ and $p_Y(y)$ respectively and joint mass function $p(x, y)$. Suppose that X and Y are independent. Then

$$\begin{aligned} E[XY] &= \sum_{(x,y) | p(x,y) > 0} x y p(x, y) \\ &= \sum_{(x,y) | p_X(x)p_Y(y) > 0} x y p_X(x)p_Y(y) \\ &= \sum_{x | p_X(x) > 0} \sum_{y | p_Y(y) > 0} x y p_X(x)p_Y(y) \\ &= \sum_{x | p_X(x) > 0} x p_X(x) \sum_{y | p_Y(y) > 0} y p_Y(y) \\ &= E[X] E[Y]. \end{aligned}$$

Example 1. This is an extended example. We start with two independent random variables X and Y that are uniformly distributed on $[0, 1]$.

So

$$f_X(x) = \begin{cases} 1 & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases} \quad \text{and} \quad f_Y(y) = \begin{cases} 1 & 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

and

$$f(x, y) = f_X(x)f_Y(y) = \begin{cases} 1 & 0 \leq x, y \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

$$E[Y] = E[X] = \int_{-\infty}^{\infty} x f_X(x) dx = \int_0^1 x \cdot 1 dx = \frac{1}{2}.$$

$$E[Y^2] = E[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x) dx = \int_0^1 x^2 \cdot 1 dx = \frac{1}{3}.$$

and so

$$\text{Var}(Y) = \text{Var}(X) = E[X^2] - (E[X])^2 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}.$$

Since X and Y are independent,

$$E[XY] = E[X]E[Y] = \frac{1}{2} \frac{1}{2} = \frac{1}{4}.$$

We now determine the probability density function for $X + Y$. We consider several cases.

Case 1. If $a \leq 0$, then $F(a) = P(X + Y \leq a) = 0$.

Case 2. If $a \geq 2$, then $F(a) = P(X + Y \leq a) = 1$.

Case 3. Let $0 \leq a \leq 1$ and let $D_a = \{(x, y) | x + y \leq a\}$, then

$$F(a) = P(X + Y \leq a) = \int \int_{D_a} f(x, y) dy dx = \int \int_{D_a \cap \text{sq}} 1 dy dx = \frac{1}{2} a^2.$$

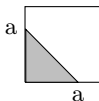


Figure 1. Area of integration for Case 3.

Case 4. Let $1 \leq a \leq 2$, and again let $D_a = \{(x, y) | x + y \leq a\}$, then

$$F(a) = \int \int_{D_a} f(x, y) dy dx = \int \int_{D_a \cap \text{sq}} 1 dy dx = 1 - \frac{1}{2}(2-a)^2.$$

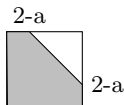


Figure 2. Area of integration for Case 4.

So in summary

$$F(a) = \begin{cases} 0 & a < 0 \\ \frac{1}{2}a^2 & 0 \leq a \leq 1 \\ 1 - \frac{1}{2}(2-a)^2 & 1 \leq a \leq 2 \\ 1 & a > 2 \end{cases}$$

and so since F is differentiable,

$$f(a) = F'(a) = \begin{cases} 0 & a < 0 \\ a & 0 \leq a \leq 1 \\ 2 - a & 1 \leq a \leq 2 \\ 0 & a > 2. \end{cases}$$

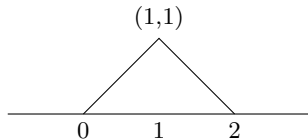


Figure 3. Graph of $f(a)$, a B-spline of order 2.

The other variable that we are interested in for this example is $X - Y$. So what is its distribution. We could use the same techniques, but let's use a trick instead.

First of all we notice that the distribution of $Y - \frac{1}{2}$ is the uniform distribution of $[-\frac{1}{2}, \frac{1}{2}]$ and that it is symmetric about zero.

This implies that the distribution for $\frac{1}{2} - Y$ is the same as the distribution of $Y - \frac{1}{2}$.

So we first observe that

$$X - Y = (X - \frac{1}{2}) - (Y - \frac{1}{2}) = (X - \frac{1}{2}) + (\frac{1}{2} - Y),$$

which has the same distribution as

$$(X - \frac{1}{2}) + (Y - \frac{1}{2}) = X + Y - 1.$$

So the graph of the pdf for $X - Y$ is

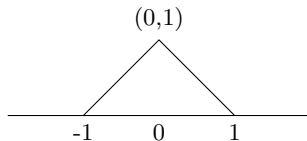


Figure 4. Graph of the pdf for $X - Y$.

We now compute $\text{Cov}(X + Y, X - Y)$.

Note first that

$$E[X + Y] = E[X] + E[Y] = \frac{1}{2} + \frac{1}{2} = 1$$

and

$$E[X - Y] = E[X] - E[Y] = \frac{1}{2} - \frac{1}{2} = 0.$$

So

$$\begin{aligned}\text{Cov}(X + Y, X - Y) &= E[(X + Y - 1)(X - Y - 0)] \\ &= E[X^2 - XY + YX - Y^2 - X + Y] \\ &= E[X^2] - E[XY] + E[YX] - E[Y^2] - E[X] + E[Y] \\ &= E[X^2] - E[Y^2] - E[XY] + E[YX] - E[X] + E[Y] \\ &= 0\end{aligned}$$

because all of the terms cancel.

So $X + Y$ and $X - Y$ are not correlated. Does this imply that they are independent? We will look into this carefully later, but note that in general, uncorrelated does not imply independence unless the variables have a joint multivariate normal distribution.

Theorem. Let $X_1, X_2, \dots, X_n$ be random variables. Then

$$\begin{aligned}\operatorname{Var}\left(\sum_{i=1}^n X_i\right) &= \operatorname{Cov}\left(\sum_{i=1}^n X_i, \sum_{j=1}^n X_j\right) \\&= \sum_{i=1}^n \sum_{j=1}^n \operatorname{Cov}(X_i, X_j) \\&= \sum_{i=1}^n \operatorname{Var}(X_i) + \sum_{i=1}^n \sum_{j \neq i}^n \operatorname{Cov}(X_i, X_j) \\&= \sum_{i=1}^n \operatorname{Var}(X_i) + 2 \sum_{i=1}^n \sum_{j < i}^n \operatorname{Cov}(X_i, X_j).\end{aligned}$$

And so if X_i and X_j are uncorrelated for each $i \neq j$, then

$$\operatorname{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \operatorname{Var}(X_i).$$

This result does not require that $X_1, X_2, \dots, X_n$ are independent, only that they are uncorrelated.

Example 2. Suppose the for any n that $X_1, X_2, \dots, X_n$ are independent random variables with mean μ and variance σ^2 (or equivalently standard deviation σ .)

Then

$$E \left[\frac{1}{n} \sum_{i=1}^n X_i \right] = \frac{1}{n} \sum_{i=1}^n E[X_i] = \frac{1}{n} \sum_{i=1}^n \mu = \mu$$

and because of the independence

$$\text{Var} \left(\frac{1}{n} \sum_{i=1}^n X_i \right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{1}{n^2} \sum_{i=1}^n \sigma^2 = \frac{\sigma^2}{n}.$$

Notice in particular that

$$\text{Var} \left(\frac{1}{n} \sum_{i=1}^n X_i \right) \rightarrow 0$$

as $n \rightarrow \infty$.

We will show later that if $X_1, X_2, \dots, X_n$ are also normally distributed, then so is $\frac{1}{n} \sum_{i=1}^n X_i$ and with mean μ , variance σ^2/n and standard deviation $\sigma/\sqrt{n}$.